10 In this question you must show detailed reasoning.

The equation of a curve is

$$y = \frac{\sin 2x - x}{x \sin x}.$$

(a) Use the small angle approximation given in the list of formulae on pages 2–3 of this question paper to show that

$$\int_{0.01}^{0.05} y \, dx \approx \ln 5. \quad \textbf{[4]}$$

(b) Use the same small angle approximation to show that

$$\frac{dy}{dx} \approx -10000 \text{ at the point where } x = 0.01. \quad \textbf{[2]}$$

The equation $y = 0$ has a root near $x = 1$. Joan uses the Newton-Raphson method to find this root. The output from the spreadsheet she uses is shown in Fig. 10.1.

n	0	1	2	3	4	5	6	7
x_n	1	0.958509	0.950084	0.948261	0.94786	0.947772	0.947753	0.947748

Fig. 10.1

Joan carries out some analysis of this output. The results are shown in Fig. 10.2.

x	y
0.9477475	–7.79967E–07
0.9477485	–2.90821E–06
x	y
0.947745	4.54066E–06
0.947755	–1.67417E–05

Fig. 10.2

(c) Consider the information in Fig. 10.1 and Fig. 10.2.

- Write 4.54066E–06 in standard mathematical notation.
- State the value of the root as accurately as you can, justifying your answer. **[3]**